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# DX Cluster Aggregator

for macOS

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## User Manual

Version 1.6.0

Aggregate FT8/FT4 spots from multiple WSJT-X/JTDX instances  
and DX Cluster nodes into a unified telnet cluster server

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### **Original Windows Application**

Vinod VU3ESV / LB9KJ

### **macOS Version Conceptualised by**

Manoj VU2CPL

### **macOS Version Developed with**

Claude Code (Anthropic)

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# Table of Contents

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- 1. Introduction**
- 2. System Requirements**
- 3. Installation**
  - 3.1 From Pre-built App Bundle
  - 3.2 Building from Source
  - 3.3 Creating the .app Bundle
- 4. Getting Started**
  - 4.1 Main Window Overview
  - 4.2 Setting Your Callsign
- 5. Configuring UDP Sources (WSJT-X / JTDX)**
  - 5.1 Adding a UDP Source
  - 5.2 Multiple Instances
- 6. Configuring DX Cluster Sources**
  - 6.1 Adding a DX Cluster Node
  - 6.2 Authentication
- 7. Broadcast Destinations**
  - 7.1 UDP Broadcast
  - 7.2 TCP Telnet Cluster Server
- 8. Monitoring & Spot Aggregation**
  - 8.1 Starting Monitoring
  - 8.2 CQ-Only Filter
  - 8.3 New-Only Filter
  - 8.4 Understanding the Spots Table
- 9. ClubLog Integration & DX Alerts**
  - 9.1 Required Credentials
  - 9.2 Configuration and Refresh
  - 9.3 Band Filter for Import
  - 9.4 Include Unconfirmed
  - 9.5 Alert Types and Highlighting
- 10. Notifications (Telegram & macOS)**
  - 10.1 Telegram Setup
  - 10.2 macOS Notifications
  - 10.3 Cooldown
  - 10.4 Notify-On Selection
- 11. Connecting Logging Software**
- 12. Troubleshooting**
- 13. Credits & Acknowledgements**

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# 1. Introduction

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DXClusterAggregator is a macOS application designed for amateur radio operators. It collects FT8, FT4, and other digital mode spots from multiple sources and aggregates them into a unified DX cluster feed that can be consumed by logging software such as Logger32, DXLab, N1MM+, Log4OM, and others.

The application supports two types of spot sources:

- **UDP Sources** - Direct connections to WSJT-X, JTDX, or other software that broadcasts decoded FT8/FT4 spots via the WSJT-X UDP protocol
- **DX Cluster Nodes** - Telnet connections to traditional DX cluster servers (DX Spider, AR-Cluster, CC Cluster, etc.) to receive spots from the worldwide cluster network

All spots from all sources are combined and rebroadcast through a built-in telnet cluster server and/or UDP, allowing your logging software to receive a single aggregated stream of spots from multiple radios, SDRs, and cluster nodes.

This macOS version is a native Swift/SwiftUI port of the original Windows application created by Vinod VU3ESV/LB9KJ. The macOS version was conceptualised by Manoj VU2CPL.

## 2. System Requirements

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| Requirement      | Details                                        |
|------------------|------------------------------------------------|
| Operating System | macOS 14.0 (Sonoma) or later                   |
| Architecture     | Apple Silicon (M1/M2/M3/M4) and Intel          |
| Disk Space       | Less than 5 MB                                 |
| Network          | Local network access for UDP/TCP               |
| Software         | WSJT-X, JTDX, or compatible decoder (optional) |

## 3. Installation

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### 3.1 From Pre-built App Bundle

- Locate **DXClusterAggregator.app** in the distribution folder
- Drag it to your **/Applications** folder (or run from any location)
- On first launch, macOS may show a security warning since the app is not notarised
- Go to **System Settings > Privacy & Security** and click **"Open Anyway"**

Alternatively, open Terminal and run the following command to bypass Gatekeeper:

```
xattr -cr /path/to/DXClusterAggregator.app
```

Then right-click the app and select "Open" for the first launch. After that it will open normally with a double-click.

## 3.2 Building from Source

If you have the source code and Xcode Command Line Tools installed (install with: `xcode-select --install`):

### Step 1: Clone and build

```
git clone https://github.com/vu2cpl/DXClusterAggregator-macOS.git
cd DXClusterAggregator-macOS
swift build -c release
```

## 3.3 Creating the .app Bundle

After building from source, you can create a proper macOS .app bundle that can be launched from Finder, placed in the Dock, or copied to /Applications:

### Step 1: Create the bundle directory structure

```
mkdir -p DXClusterAggregator.app/Contents/MacOS
mkdir -p DXClusterAggregator.app/Contents/Resources
```

### Step 2: Copy the binary and icon

```
cp .build/release/DXClusterAggregator DXClusterAggregator.app/Contents/MacOS/
cp AppIcon.icns DXClusterAggregator.app/Contents/Resources/
echo -n "APPL?????" > DXClusterAggregator.app/Contents/PkgInfo
```

### Step 3: Create the Info.plist

Create the file `DXClusterAggregator.app/Contents/Info.plist` with the following content (a template is provided in the project README):

| Key                     | Value                         |
|-------------------------|-------------------------------|
| CFBundleName            | DXClusterAggregator           |
| CFBundleDisplayName     | DX Cluster Aggregator         |
| CFBundleIdentifier      | com.vu2cpl.dxcusteraggregator |
| CFBundleVersion         | 1.2.0                         |
| CFBundleExecutable      | DXClusterAggregator           |
| CFBundlePackageType     | APPL                          |
| CFBundleIconFile        | AppIcon                       |
| LSMinimumSystemVersion  | 14.0                          |
| NSHighResolutionCapable | true                          |

*The full Info.plist XML template is available in the project README on GitHub.*

### Step 4: Sign and launch

```
codesign --force --deep --sign - DXClusterAggregator.app
open DXClusterAggregator.app
```

### Step 5 (Optional): Install to Applications

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```
cp -r DXClusterAggregator.app /Applications/
```

*If sharing the built .app with others, they will need to run `xattr -cr /path/to/DXClusterAggregator.app` and right-click > Open on the first launch, as the app is not notarised through the Apple Developer Program.*

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## 4. Getting Started

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### 4.1 Main Window Overview

When you launch DXClusterAggregator, you will see the main window with the following sections from top to bottom. The **Hide Settings** button in the header collapses the entire configuration panel (Configuration, UDP Sources, DX Cluster, ClubLog) so the spots table can use the full window. Click **Show Settings** to bring the configuration back.

| Section          | Purpose                                                                 |
|------------------|-------------------------------------------------------------------------|
| Header           | App name and version number                                             |
| Configuration    | Callsign, TCP cluster port, and broadcast destinations                  |
| UDP Sources      | List of WSJT-X/JTDX UDP sources with IP, port, and status               |
| DX Cluster Nodes | List of telnet DX cluster sources with address, credentials, and status |
| Controls         | CQ-Only filter, Minimize on Start, Clear Spots, Start/Stop Monitoring   |
| Spots Table      | Real-time display of aggregated spots from all sources                  |
| Status Bar       | Connection status, active source count, and total spot count            |

### 4.2 Setting Your Callsign

Enter your callsign in the **Callsign** field at the top of the window. This callsign is used as the spotter identifier when rebroadcasting spots through the cluster server. The callsign is automatically converted to uppercase. It is also used as the default username when adding new DX Cluster sources.

## 5. Configuring UDP Sources (WSJT-X / JTDX)

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UDP Sources receive decoded FT8/FT4/JT65/JT9 spots directly from WSJT-X, JTDX, or any software that implements the WSJT-X UDP protocol. The app parses the binary WSJT-X protocol messages (Status and Decode types) to extract spot information.

### 5.1 Adding a UDP Source

Click **"Add UDP Source"** in the UDP Sources section. Configure the following fields:

| Field     | Description                                          | Default  |
|-----------|------------------------------------------------------|----------|
| Name      | A label to identify this source (e.g., "WSJT-X 20m") | Source N |
| Listen IP | IP to listen on. Use 0.0.0.0 for all interfaces      | 0.0.0.0  |
| Port      | UDP port number matching your WSJT-X/JTDX setting    | 2237     |

|         |                                      |    |
|---------|--------------------------------------|----|
| Enabled | Toggle to enable/disable this source | On |
|---------|--------------------------------------|----|

*Note: In WSJT-X, go to File > Settings > Reporting and set the UDP Server address to 127.0.0.1 (or your Mac's IP if running on another machine) and the port to match the port configured here.*

## 5.2 Multiple Instances

You can add multiple UDP sources to aggregate spots from several WSJT-X/JTDX instances running simultaneously. Each instance must be configured to broadcast on a different UDP port. For example:

- WSJT-X on 20m band: UDP port 2237
- JTDX on 40m band: UDP port 2238
- WSJT-X on 15m band: UDP port 2239

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## 6. Configuring DX Cluster Sources

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DX Cluster sources connect to traditional DX cluster telnet servers to receive spots from the worldwide amateur radio spotting network. These spots are aggregated alongside your local WSJT-X/JTDX decodes.

### 6.1 Adding a DX Cluster Node

Click **"Add DX Cluster"** in the DX Cluster Nodes section. Configure the following fields:

| Field    | Description                                             | Default       |
|----------|---------------------------------------------------------|---------------|
| Name     | A label for this cluster (e.g., "VE7CC")                | Cluster N     |
| Address  | Hostname or IP of the cluster node                      | (empty)       |
| Port     | Telnet port number                                      | 7300          |
| Username | Your callsign for login                                 | Your callsign |
| Password | Password if required (many clusters do not require one) | (empty)       |
| Enabled  | Toggle to enable/disable                                | On            |

#### Popular DX Cluster Nodes

| Node   | Address                | Port |
|--------|------------------------|------|
| VE7CC  | dxc.ve7cc.net          | 23   |
| W9PA   | dxc.w9pa.net           | 7300 |
| GB7DXC | gb7dxc.dxcluster.co.uk | 7300 |
| WA9PIE | dxc.wa9pie.net         | 7300 |
| K1TTT  | k1ttt.net              | 7300 |

### 6.2 Authentication

The application automatically detects login and password prompts from the cluster server. Most DX cluster nodes only require a callsign (no password). The app sends your username when it detects a login prompt, and your password (if provided) when it detects a password prompt. Connection status is shown next to each cluster entry:

- **Connecting...** - TCP connection is being established
- **Connected** - TCP connection established, waiting for login
- **Authenticated** - Successfully logged in and receiving spots

## 7. Broadcast Destinations

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Aggregated spots are rebroadcast through two mechanisms:



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## 7.1 UDP Broadcast

Two UDP broadcast destinations can be configured. Each has an IP address and port. Spots are sent as DX cluster-formatted text lines via UDP. Click **Save** after changing the settings. These can be used to feed spots to other applications on the local machine or across the network.

## 7.2 TCP Telnet Cluster Server

The app runs a built-in TCP telnet cluster server. Any logging software that supports connecting to a DX cluster via telnet can connect to this server to receive the aggregated spot feed. The default port is **7550**.

Configure your logging software to connect to:

- **Host:** 127.0.0.1 (if on the same machine) or your Mac's IP address
- **Port:** 7550 (or whatever you configured)

The status bar shows the number of connected telnet clients.

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## 8. Monitoring & Spot Aggregation

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### 8.1 Starting Monitoring

Click the green **"Start Monitoring"** button to begin. This will:

- Start a UDP listener for each enabled UDP source
- Connect to each enabled DX Cluster node
- Start the TCP telnet cluster server
- Configure the UDP broadcast destinations

*While monitoring is active, source settings cannot be modified. Click the red **"Stop Monitoring"** button to stop and reconfigure.*

### 8.2 CQ-Only Filter

Enable the **CQ Only** toggle to filter spots so that only CQ calls are displayed and rebroadcast. This reduces the volume of spots to only stations calling CQ, which are the ones available to work. QSO exchanges (reports, RRR, 73) are filtered out.

### 8.3 New-Only Filter

Enable the **New Only** toggle to filter spots so that only those matching an enabled ClubLog alert are displayed and rebroadcast. A spot passes the filter if its classification is New DXCC, New Slot, New Band, or New Mode (whichever you have enabled in the ClubLog section). Worked stations and unclassified spots are hidden. This filter requires ClubLog data to be loaded via Refresh - without log data, every spot is unclassified and nothing will be shown.

*Tip: combine **CQ Only** + **New Only** for the cleanest possible feed - only stations calling CQ that you actually need.*

### 8.4 Understanding the Spots Table

The spots table displays all aggregated spots in real-time:

| Column     | Description                                        |
|------------|----------------------------------------------------|
| Time       | UTC time of the spot (HHmm format)                 |
| Source     | Name of the source that provided this spot         |
| Callsign   | The DX station callsign extracted from the message |
| Freq (MHz) | Frequency in MHz (dial frequency + audio offset)   |
| SNR        | Signal-to-noise ratio in dB (from WSJT-X decodes)  |
| Mode       | Operating mode (FT8, FT4, CW, SSB, etc.)           |
| Message    | Full decoded message or spot comment               |

*CQ spots are highlighted with a green background in the table for easy identification.*

## 9. ClubLog Integration & DX Alerts

The ClubLog integration (new in v1.3.0) downloads your personal log from ClubLog and classifies each incoming spot as new DXCC, new slot, new band, new mode, or already worked. Spots are color-highlighted in the spots table so you can instantly see which stations are worth working.

### 9.1 Required Credentials

You will need the following from your ClubLog account (free at <https://clublog.org>):

| Field        | What it is                                     | Where to get it                                                                           |
|--------------|------------------------------------------------|-------------------------------------------------------------------------------------------|
| Callsign     | Your amateur radio callsign                    | Your ClubLog account callsign                                                             |
| Email        | Email address registered with ClubLog          | Your ClubLog account email                                                                |
| App Password | Token for API access (NOT your login password) | ClubLog > Settings > App Passwords                                                        |
| API Key      | Developer key for country file download        | <a href="https://clublog.org/requestapikey.php">https://clublog.org/requestapikey.php</a> |

*All credentials are stored locally on your Mac in UserDefaults. They are not transmitted anywhere except to ClubLog's own servers when you click Refresh.*

### 9.2 Configuration and Refresh

In the ClubLog Integration section:

- Enter your **Callsign**, **Email**, **App Password**, and **API Key**
- Select which **Alert Types** you want highlighted (New DXCC, New Slot, New Band, New Mode)
- Click **Refresh from ClubLog** - this downloads the country file and your full ADIF log
- Wait for the status message to show the QSO and DXCC count (depending on log size, this may take 30-120 seconds)

The downloaded data is cached locally at `~/Library/Application Support/DXClusterAggregator/` so the app loads instantly on next launch without re-downloading. You should refresh periodically (e.g. after each logging session) to keep the worked-status data up to date.

### 9.3 Band Filter for Import

The **Import Bands** grid in the ClubLog section lets you choose which bands' QSOs are imported from your ClubLog log. Useful if you only operate certain bands and want a smaller, faster matrix.

- **All** button: import every band (default)
- **HF Only** button: 160M through 10M
- Individual band checkboxes: pick exactly the bands you want

*Changes take effect on the next Refresh (the import filter is applied during ADIF parsing).*

### 9.4 Include Unconfirmed

Enable the **Inc. Unconfirmed** toggle to make worked-but-unconfirmed slots count as still needed. A slot is considered confirmed if any of LOTW\_QSL\_RCVD, QSL\_RCVD, or EQSL\_QSL\_RCVD is set to Y in the ADIF record (also recognises ClubLog's matched flag). When this toggle is ON, the alert classifier

compares against your confirmed slots only, so QSOs awaiting confirmation will trigger New DXCC / New Slot / etc. alerts. This is useful for award-chasers who want to re-work stations until confirmation is received.

## 9.5 Alert Types and Highlighting

A **slot** means a specific DXCC + Band + Mode combination. For example, if you have worked Japan on 20M FT8 and 40M CW, then a spot of a Japanese station on 15M FT8 is a **new slot** - Japan is worked (so not new DXCC), 15M is worked (for other entities), FT8 is worked, but this specific JA+15M+FT8 combination is not. Slots matter for awards like 5BDXCC, 9BDXCC, and triple-play.

| Indicator      | Level    | Meaning                                                                |
|----------------|----------|------------------------------------------------------------------------|
| ■ red row      | New DXCC | You have never worked this DXCC entity                                 |
| ■ orange row   | New Slot | You have worked this DXCC but not on this exact band+mode combination  |
| ■ blue row     | New Band | You have worked this DXCC but not on this band (any mode)              |
| ■ amber row    | New Mode | You have worked this DXCC but not in this mode (any band)              |
| ■ no highlight | Worked   | You have already worked this station/slot - not needed                 |
| (blank)        | Unknown  | Classification not possible (DXCC not resolved, or no log data loaded) |

Priority order: New DXCC beats New Slot beats New Band beats New Mode beats Worked. If you disable a specific alert type via the toggle, spots at that level are displayed as Worked (no highlight).

The spots table now includes two extra columns: **DXCC** (the entity name, e.g. "UNITED STATES") and **Band** (e.g. "20M"). A small colored indicator appears at the far-left of each row showing the alert level at a glance.



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## 10. Notifications (Telegram & macOS)

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DXClusterAggregator can push alerts to your Telegram chat and/or to the macOS Notification Center whenever a spot matching your selected alert types arrives. Alerts are throttled per callsign by a configurable cooldown to prevent spam.

### 10.1 Telegram Setup

- Create a Telegram bot via @BotFather and copy the **Bot Token**
- Send /start to your bot, then look up your **Chat ID** via @userinfobot or by visiting <https://api.telegram.org/bot<TOKEN>/getUpdates>
- In DXClusterAggregator, open the **Notifications** section and enable **Telegram**
- Paste the Bot Token and Chat ID, then click **Send Test**

### 10.2 macOS Notifications

Enable the **macOS Notifications** toggle. The first time you enable it, macOS will prompt you to allow notifications from DXClusterAggregator. Click **Allow**. Banners will then appear in the corner of your screen for each matching spot.

*If you accidentally denied permission, open System Settings > Notifications, find DXClusterAggregator in the list, and re-enable Allow Notifications.*

### 10.3 Cooldown

Set the **Cooldown** to a value between 5 and 60 minutes (default 15). Within this window, repeat spots of the same callsign do not trigger another notification. This prevents getting flooded by FT8 stations that decode every 15 seconds.

### 10.4 Notify-On Selection

Use the **Notify on** checkboxes to choose which alert levels should trigger a notification - independent of the table-highlight toggles in the ClubLog section. For example, you might highlight all four levels visually but only get notifications for new DXCC and new slot.

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# 11. Connecting Logging Software

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DXClusterAggregator acts as a telnet DX cluster server. To connect your logging software:

## Logger32

- Go to Setup > Telnet > DX Cluster
- Set Host to **127.0.0.1** and Port to **7550**
- Click Connect

## N1MM+

- Go to Window > Telnet
- Enter **127.0.0.1:7550** as the cluster address

## Log4OM

- Go to Settings > Cluster
- Add a new cluster with address **127.0.0.1** port **7550**

## Any Telnet Client

You can also test the cluster server using a terminal:

```
telnet 127.0.0.1 7550
```

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## 12. Troubleshooting

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### **No spots appearing from WSJT-X**

Verify the UDP port in WSJT-X Settings > Reporting matches the port configured in DXClusterAggregator. Ensure WSJT-X is set to broadcast to 127.0.0.1 (or your Mac's IP). Check that the source is enabled and shows "Active" status.

### **DX Cluster not connecting**

Verify the cluster address and port are correct. Check your internet connection. Some clusters may be down - try a different node. Check if a firewall is blocking outbound TCP connections.

### **DX Cluster shows "Waiting..."**

The cluster node may be unreachable or the DNS lookup failed. Try using an IP address instead of a hostname.

### **macOS security warning on launch**

Go to System Settings > Privacy & Security and click "Open Anyway". This is needed because the app is not notarised through the Apple Developer Program.

### **Logging software cannot connect**

Ensure monitoring is started (green "Start Monitoring" button). Verify the TCP cluster port (default 7550) is not blocked by a firewall. Check that the port is not already in use by another application.

### **Port already in use**

Another application is using the same port. Either close that application or change the port in DXClusterAggregator to an unused port number.

### **Spots from cluster but no frequency**

DX Cluster spots include frequency information from the cluster server. This is normal - the frequency comes from the spotter, not from your radio.



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## 13. Credits & Acknowledgements

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### Original Windows Application

The original DXClusterAggregator was developed as a Windows .NET application by **Vinod, VU3ESV / LB9KJ**. His vision of aggregating FT8 spots from multiple sources into a unified DX cluster feed laid the foundation for this project.

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### macOS Version

The macOS version of DXClusterAggregator was conceptualised by **Manoj, VU2CPL**. This native Swift/SwiftUI port brings the functionality of the original Windows application to the Mac platform, with enhancements including multiple UDP source support, DX Cluster telnet node integration, and a modern macOS user interface.

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### Development

The macOS application was developed with the assistance of **Claude Code** by Anthropic, an AI-powered software engineering tool.

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### Open Source Libraries & Protocols

- **WSJT-X UDP Protocol** - Joe Taylor K1JT and the WSJT-X development team
- **DX Cluster Protocol** - DX Spider and AR-Cluster communities
- **Apple Network Framework** - For TCP/UDP networking on macOS
- **SwiftUI** - Apple's modern UI framework for macOS

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73 de VU2CPL

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